This project was carried out with the aid of AI (gpt 5.6).

Folding

In this lecture, k = alg closed field, R = smooth finite dimensional ordinary algebra.

Let $\Delta = k[\varepsilon]/\varepsilon^2$, $R_\Delta = R \otimes_k \Delta$

Goal: Lemma:

We have $Sg_{dg}(R_\Delta) \cong Perf(Perf(R)/[1])$

$\cong Perf(Perf(R) \otimes_k 2\ell)$.

Here $2\ell = k[t, t^{-1}]$, $|t| = 1$, DG structure: $d = 0$.

Let $MF\text{-}mod_{dg}(R)$. Forget grading:

$\bigoplus_{i \in \mathbb{Z}} M^i \in mod(R_\Delta)$. $R_\Delta = R \otimes_k \Delta$

$x(a \otimes 1) = xa$, $x(1 \otimes \varepsilon) = dx$, (note that $d^2 = 0$)

Denote this by $Fold(M) \in mod(R_\Delta)$

$C_1(mod(R))$ = cat of 1-periodic complexes

Objects: (N, ∂), $N \in mod(R)$, $\partial : N \to N$ R-linear, with $\partial^2 = 0$.

$$\mathrm{Hom}_{C_1(\mathrm{mod}(R))}\big((M,\partial_M),(N,\partial_N)\big) = \{f \in \mathrm{Hom}_R(M,N) : \partial_N f = f \partial_M\}.$$

this is an abelian category.

$$\mathrm{Tot}_1 : C^b(\mathrm{mod}(R)) \longrightarrow C_1(\mathrm{mod}(R))$$
$$(M^\cdot, d) \longmapsto \Big(\bigoplus_{i\in\mathbb{Z}} M^i,\ \partial = d\Big).$$

$$\begin{array}{ccc} C^b(\mathrm{mod}(R)) & & \\ \downarrow \mathrm{Tot}_1 & \searrow \text{forget} & \\ C_1(\mathrm{mod}(R)) & \xrightarrow[\text{equiv of abelian cat.}]{\sim} & \mathrm{mod}(R_\Delta) \end{array}$$

In C_1: homotopic morphisms are defined.

Modulo those: $K_1(\mathrm{mod}(R))$

inverting quasi-iso: $D_1(\mathrm{mod}(R))$.

Motivation:

Recall: Def: A finite dim alg A is self-injective if A_A is injective.

$\mathrm{mod}(A)$ is Frobenious cate.

$\underline{\mathrm{mod}}(A)$: $\mathrm{Obj} = \mathrm{Obj}(\mathrm{mod}\, A)$

$$\underline{\mathrm{Hom}}_A(X,Y) = \mathrm{Hom}_{\underline{\mathrm{mod}}(A)}(X,Y)$$
$$= \mathrm{Hom}_A(X,Y) \Big/ \{\text{maps factoring through finite gen projective modules}\}$$

mod(A) is triangulated.

Theorem: Rickard: Derived cat and stable eq 1989.

Let A finite dim self-inj alg.

$\underline{mod}(A) \simeq D_{sg}(A)$ triangulated equivalence.

Def: A 2-sided Noetherian algebra A is Iwanaga-Gorenstein if $\mathrm{inj\,dim}(A_A) < \infty$, $\mathrm{inj\,dim}({}_AA) < \infty$.

Def: A finitely generated A-module G is Gorenstein projective, if it is a ker module of a totally acyclic complex of finite gen projectives

$$\cdots \to P^{-1} \to P^0 \to P^1 \to \cdots$$

the complex is exact, remains exact under $\mathrm{Hom}(-, Q)$ for every projective Q. Write $G\mathrm{proj}(A)$ for Cat of G proj A-module.

Lemma: For a selfinj A: $G\mathrm{proj}(A) = \mathrm{mod}(A)$

For a smooth A: $G\mathrm{proj}(A) = \mathrm{proj}(A)$.

$\underline{G\mathrm{proj}}(A)$ = same obj as $G\mathrm{proj}(A)$

$\underline{\mathrm{Hom}}$ = Hom / maps factor through proj

Theorem [Buchweitz: Maximal CM modules ... 1987.]

If A is a finitedim Iwanaga–Gorenstein algebra, then $D_{sg}(A) \simeq \underline{G\text{proj}}(A)$

Lemma: Let R smooth finite dim: then R_Δ is Iwanaga–Gorenstein. So

$$D_{sg}(R_\Delta) \simeq \underline{G\text{proj}}(R).$$

pf: $R \subseteq R_\Delta$ is a Frobenius inclusion: i.e.

$$R_\Delta \cong \mathrm{Hom}_R((R_\Delta)_R, R)$$

Res-Coind adjunction: R_Δ

$$\mathrm{Hom}_{R_\Delta}(N, \mathrm{Hom}_R(R_\Delta, R)) \cong \mathrm{Hom}_R(N, R)$$

$$\mathrm{Ext}^i_{R_\Delta}(N, R_\Delta) \cong \mathrm{Ext}^i_R(N, R).$$

$g\dim(R) = d < \infty$ so these vanish for $i > d$.

Similar for left modules. $\Rightarrow$ $i\text{-}\dim(R_\Delta)$ as right and left modules are finite. □

Theorem [Jiaqun Wei: Grostein homological theory ... 2015]

Assume R has finite global dim. We have.

$$G\text{proj}(R_\Delta) \cong C_1(\text{proj}(R)).$$

Lemma: Under above identification:

projective R_Δ-modules are exactly the contractible objects in $C_1(proj(R))$. Hence we have

Wei: $$\boxed{G_{proj}(R_\Delta) \cong K_1(proj(R))}$$

Proof: Take the free R_Δ-module R_Δ, correspond to $(R\oplus R, \partial)$, $\partial = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$

Let $s = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, then $s\partial + \partial s = 1$.

So this is contractible, and projective R_Δ-modules are hence contractible.

Conversely, take $(P, \partial) \in C_1(proj(R))$

Contractible: choose s: $s\partial + \partial s = 1$,

So $\partial s \partial = \partial$, ∂s is idempotent.

Set $Q = Im(\partial s)$, so $Q \in proj(R)$.

Check: $P \cong Q \oplus s(Q)$, $\partial = s(Q) \to Q$, $\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$

So $P = Q \otimes_R R_\Delta$, it is projective over R_Δ □

Lemma [Xiaojun Zhao, A note on the eq of m-per ---- 2014]

$$K_1(proj(R)) \cong D_1(mod(R))$$

Lemma: We have

$$D_{sg}(R_1) \cong D_1(\mathrm{mod}(R))$$

Proof: Buchweitz: $D_{sg}(R_0) \cong \underline{G\mathrm{proj}}(R)$.

Wei: $\underline{G\mathrm{proj}}(R) \cong K_1(\mathrm{proj}(R))$

Zhao: $K_1(\mathrm{proj}(R)) \cong D_1(\mathrm{mod}(R))$. □

Observe:

$$\mathrm{Hom}_{D_1(\mathrm{mod}R)}(F_0\mathrm{Id}(X), F_0\mathrm{Id}(Y))$$

$$= \bigoplus_{j\in\mathbb{Z}} \mathrm{Hom}_{D(R)}(X, Y[j]).$$

Let $\mathcal{C}$ = pretriangulated DG cate.

orbit category: $\mathcal{C}/[1]$, $\mathrm{obj} = \mathrm{obj}\,\mathcal{C}$,

$$\mathrm{Hom}_{\mathcal{C}/[1]}(X,Y) = \bigoplus_{j\in\mathbb{Z}} \mathrm{Hom}_{\mathcal{C}}(X, Y[j])$$

[Keller: On triangulate orbit cate 2005]

Take pretriangulate hull:

$$H^0(\mathrm{Perf}(\mathrm{Perf}(R)/[1])) \cong D_1(\mathrm{mod}(R))$$

Let $\mathcal{U} = k[t,t^{-1}]$, $|t|=1$, $d=0$.

$\mathcal{C} \otimes_k \mathcal{U}$. $\mathrm{obj} = \mathrm{obj}(\mathcal{C})$.

$$\mathrm{Hom}_{\mathcal{C}\otimes_k\mathcal{U}}(X,Y) = \mathrm{Hom}_{\mathcal{C}}(X,Y) \otimes_k \mathcal{U}.$$

$$\simeq \bigoplus_{j\in\mathbb{Z}} \mathrm{Hom}_{\mathcal{C}}(X, Y[j])$$

Can check: $\mathcal{C} \otimes_k \mathcal{U} \simeq_{dg} \mathcal{C}/[1]$.

So $H^0(\mathrm{Perf}(R_1) \otimes \mathcal{U}) = D_s(\mathrm{mod}(R))$.

(work): Lemma:

$$Sg_{dg}(R_1) \simeq \mathrm{Perf}(\mathrm{Perf}(R) \otimes_k \mathcal{U}).$$

$\mathcal{P}$, phantom $= \mathrm{Perf}(\Gamma) / \langle L_1, \cdots, L_{13} \rangle$

Can we arrange the quotient on ? ?